Aziz Shameem

azizshameem.github.io | Aziz-Shameem | azíz-shameem

Education

Indian Institute of Technology Bombay — Mumbai, India
B.Tech in Electrical Engineering and M.Tech in Artificial Intelligence ***(Cumulative GPA: 9.72/10.0)*** — *Nov'20 – Aug'25*

Scholastic Achievements

- Recipient of the **Institute Silver Medal** from IIT Bombay, for outstanding academic performance overall
- Ranked **1st** in **Electrical Engineering Dept.** and **Centre for Machine Intelligence and Data Science** (*2022-25*)
- Recipient of the **Summer@EPFL Travel Scholarship** (awarded to 1.6% of the applicants) (*2024*)
- Awarded **Institute Academic Prize** for exceptional performance in academics for **two consecutive years** (*2022, 2023*)
- Secured an **AIR of 62** in JEE Mains (Engineering) and an **AIR of 665** in JEE Advanced (*2020*)
- Conferred with **AP grade (Advanced Performer)** in **Advanced ML, Linear Algebra, Electronic Devices**

Publications

- M. Shukla, **A. Shameem**, M. Salzmann, A. Alahi, "**Towards Self-Supervised Covariance Estimation in Deep Heteroscedastic Regression**", ICLR 2025
- V. Prasad, **A. Shameem**, C. White, S. Nayak, P. Jain, P. Garg, G. Ramakrishnan, "**Speeding up NAS with Adaptive Subset Selection**", AutoML 2024

Research Experience

Variational Auto-Encoders and Diffusions Models — *May'24 – Dec'24*
Summer Internship, Guide: Prof. Alexandre alahi, Prof. Amit Sethi — *EPFL, EE IITB*

- Performed extensive analysis of the **similarities and differences** between two types of Generative Frameworks - **Variational AutoEncoders** and **Diffusion Models**
- Designed experiments to elucidate the **limitations** of VAEs, and suggested **improvements/methods** to overcome them
- Demonstrated methods for better generations by VAEs on **synthetic** as well as **benchmark datasets**
- Incorporated the methods for improving **conditional image generation using VAEs**
- Introduced **Probabilistic Optimizations** to better the computational complexity of the algorithms

OPAS : Ordered Permutation-based Alignment for Sequence matching — *Jun'23 – Oct'24*
Guide: Prof. Abir De — *CSE, CMInDS, IIT Bombay*

- Developed a **theoretical framework** based on **lagrangians** for **fast, approximate, constrained sequence matching**
- Tested the developed framework **computationally**, and obtained results on datasets of several **modalities**
- Compared the developed method with several baselines, and demonstrated over **58x speedup** over them
- Compared several **implementation variants, architectures and models** of the proposed system as ablations

Professional Experience

Modem Systems Engineering Intern — *May'23 – Jul'23*
Qualcomm Technologies Inc. — *Bengaluru, India*

- Undertook extensive research on enhancing **information-deficit channel estimation** methods for **5G**
- Implemented existing as well as proposed technology in a **python simulator** and verified theoretical results
- Implemented specific **channel models** in the python simulator, to be used for future research and developement

Technical Skills

Data Analysis/Visualization tools : Pandas, Numpy, Matplotlib, seaborn, plotly, MATLAB, MySQL
Libraries: Pytorch, Tensorflow, Numpy, Pandas, OpenCV, SciPy, Seaborn, Scikit-Learn, OpenCV, Matplotlib
Software: Git, LaTeX, MATLAB, GNU Radio, Quartus, NGSpice, Kicad, Arduino, Pycharm, Anaconda, Unity

Key Coursework

Machine Learning : Advanced ML, Deep Learning for NLP, Learning with Graphs, Mathematical Optimization Techniques, Optimization in ML, Online Learning and Optimization, Introduction to Machine Learning
Computer Science : Programming for Data Science, Image Processing, Design and Analysis of Algorithms, Data Structures and Algorithms, Digital Systems, Microprocessors, Computer Programming and Utilization
Math & Probability : Markov Chains and Queueing Processes, Applied Linear Algebra, Advanced Probability and Random Processes, Matrix Computations, Estimation and Identification

Key Technical Projects

Auto-Grad Engine and GPT developement *May'23 – Jul'23*
Self Project *IIT Bombay*

- Built an **Auto-Grad Engine** having **pytorch-like API** from scratch and demonstrated its working on a test bed
- Preformed extensive study on **innate problems** in neural networks pertaining to **gradients and dead neurons**
- Implemented initialization and normalization based methods to bypass problems and demonstrated better results
- Developed a working **GPT-like architecture** from scratch and generated recognizable **Shakespearean text**

Toxicity Detection in Large Language Models *Apr'24 – May'24*
Guide: Prof. Pushpak Bhattacharya(CS772 : DL for NLP) *IIT Bombay*

- Trained a **text toxicity classifier** using **Recurrent Neural Nets**, **LSTMs** and **Transformers**
- **Engineered additional features** from the analysis of textual data in an effort to improve classification of toxicity in texts
- Utilised the trained classifier network as a **filter** to mitigate toxicity in the output of **Large Language Models**

Correlated Multi-Armed Bandits *Mar'23 – Apr'23*
Guide: Prof. Jayakrishnan Nair, Prof. D. Manjunath (EE6106: Online Learning and Optimization) *IIT Bombay*

- Performed an extensive literature review on the formulation of the **correlated multi-armed Bandit** problem
- Applied an existing framework for the **exploitation of correlation** between bandits, on different algorithms and demonstrated **substantial improvements** in the same

Music Genre Classification using ML and DL *Apr'22 – may'22*
Guide: Prof. Biplab bannerjee(DS303: Intorduction to Machine Learning) *IIT Bombay*

- Implemented **Decision Trees**, **Random Forest**, **Naive Bayes**, **SVM** and **KNN** to classify music based on its genre
- Tried several architectures and hyper-parameters of **Sequential Neural Networks** to classify music based on genre

Gas Leakage Detection and VSLAM using Nanosaur *Jan'23 – Apr'23*
Guide : Prof. Siddharth Tallur (EE344 : Electronic Design Lab) *IIT Bombay*

- Ideated and designed a mobile robot capable of **toxic gas detection** and **3D reconstruction** of its environment
- Carried out the design and testing of the **Printed Circuit Board** for interfacing the gas sensors with Jetson Nano
- Created **ROS Nodes** for efficient **sensor data accumulation and display**, and got **recognized** for the same

Positions of Responsibility

Machine Learning Engineer
Data Analysis and Visualisation Team, IITB *Jun'22 – Apr'23*

- Led a team to **successful completion and publication** (On the official public handle) of analysis on **Indian Establishments and Settlements**, **Grading Statistics** and **Semester Exchange statistics**
- Implemented the **YOLO version-1** framework from scratch as part of a study project and gave a presentation on the same
- Delivered a comprehensive presentation on **Loss Functions for Image Restoration** and got recognized for the same

Mentor - Introduction to Machine Intelligence
Seasons of Code, IITB *Apr'22 – Jul'22*

- Prepared and distributed **comprehensive resources** to help beginners understand and appreciate concepts in ML
- Assisted beginners in building **AI systems** using **Reinforcement Learning** in an effort to generate interest in the field

Web Developer/Coordinator
Techfest - Asia's largest science and technology festival *Oct'21 – Feb'22*

- Contributed to the creation of **TechFest Website** by developing **hover animations, Bootstrap cards and Carousels**
- Improvised and optimised existing code-base by appropriate fixes and restructuring

Miscellaneous

TAship	Complex Analysis; Introduction to Machine Learning; Linear Algebra; Programming for Data Science
Others	Served as the **Class Representative** for the CMInDS IDDDP batch of 2020 Stood **third** in an **Algorithmic Trading hackathon** organised by **Web and Coding club, IITB** Secured **First Place** in **9-Up**, a chess tournament organized by Hostel-9 sports council Represented **Hostel 9** in **Tennis, Chess** at the **General Championships** at IITB for two consecutive years